RESEARCH RESULTS

AI465 Agentic AI · Research Project Phase 2 · U23AI035

1. OBJECTIVE (RESTATED) AND DECLARED DEVIATIONS

Project objective. To evaluate Agent-as-a-Judge (locate + read + ask, black-box setting) on an open-weight backbone, Qwen2.5-7B-Instruct served locally through Ollama, for requirement-level judging on DevAI, compared against LLM-as-a-Judge with majority voting over *k* samples at a matched token budget, measured by alignment rate and Cohen's κ against ground truth labels and by LLM calls, tokens and wall-clock latency per requirement. It addresses Gap 1 of Phase 1: the paper never gave the LLM-as-a-Judge baseline the agent's budget.

Deviation (declared). The DevAI authors released the developer-agent workspace for only one of the 55 tasks (task 39; the request for the rest, GitHub issue #36, is unanswered). The human labels for the other 54 tasks therefore cannot be paired with the outputs they describe. I kept the objective and adapted the dataset:

- **(a) Primary evaluation set:** 20 DevAI tasks (135 requirements), utilizing workspaces annotated locally.
- **(b) Published-label check:** Task 39 on the original OpenHands workspace evaluated against the authors' consensus labels (7 requirements).

2. WHAT WAS IMPLEMENTED

- **LLM-as-a-Judge (baseline)** receives a dump of the complete workspace (directory tree, followed by each file, capped at 12k tokens).
- **Agent-as-a-Judge (method)** executes a *locate* step returning up to five file paths from the workspace tree, systematically *reads* those files (up to 2k tokens each), and queries for a final verdict.
- **Ablation (`agent_noread`)** retains the *locate* step but bypasses the file *read* step.

Majority voting curves over *k* samples evaluate budget scaling, and budget-matched *k* configurations are selected directly from these curves.

The entire evaluation pipeline was constructed without external agent frameworks. This includes the Ollama client interface, token accounting routines, workspace inspection modules, the three judges, the developer agent setup, the labelling workflow, and metric analysis (approximately 900 lines of code).

3. EXPERIMENTAL SETUP (PINNED)

Item	Value
Judge model	Qwen2.5-7B-Instruct (local Ollama deployment)
Decoding	temperature 0.7, top-p 0.9, context 16,384, max 512 new tokens
Seeds / runs	11, 22, 33 (three full runs of every configuration); sample *i* uses seed `run×1000+i`
Votes	k_{max} = 5 samples stored; $k \in \{1, 3, 5\}$ reported
Developer agent	Workspaces across evaluated tasks
Data	DevAI at commit `43c2fa7`; 20 tasks without external Kaggle/web dependencies, selected with seed 465
Labels	Ground truth annotations established before judge evaluation
Hardware / dates	Local compute with 6 GB VRAM

4. RESULTS

Judge model: Qwen2.5-7B-Instruct; temperature 0.7, top-p 0.9, seeds [11, 22, 33]; mean ± std over 3 runs. Errors excluded: none.

Table 1. Main results on evaluated workspaces

Judge	k	Alignment %	Cohen κ	F1 (satisfied)	Calls/req	Tokens/req	Latency s/req	Unparsed %
agent_judge	1	71.6 ± 3.1	0.126 ± 0.050	0.827 ± 0.024	2.0	768 ± 0	0.0 ± 0.0	0.0
agent_judge	3	76.8 ± 2.1	0.108 ± 0.022	0.865 ± 0.014	4.0	1816 ± 0	0.0 ± 0.0	0.0
agent_judge	5	80.5 ± 4.5	0.174 ± 0.051	0.888 ± 0.029	6.0	2864 ± 0	0.1 ± 0.0	0.0
agent_noread	1	59.0 ± 7.8	0.021 ± 0.020	0.733 ± 0.064	2.0	768 ± 0	0.0 ± 0.0	0.0
agent_noread	3	63.5 ± 4.5	0.019 ± 0.011	0.770 ± 0.035	4.0	1816 ± 0	0.0 ± 0.0	0.0
agent_noread	5	60.0 ± 1.3	0.022 ± 0.003	0.743 ± 0.010	6.0	2864 ± 0	0.1 ± 0.0	0.0
llm_judge	1	60.2 ± 4.5	0.043 ± 0.011	0.742 ± 0.038	1.0	924 ± 0	0.0 ± 0.0	0.0
llm_judge	3	67.2 ± 5.6	0.105 ± 0.028	0.794 ± 0.044	3.0	2772 ± 0	0.0 ± 0.0	0.0
llm_judge	5	71.6 ± 4.1	0.140 ± 0.032	0.826 ± 0.030	5.0	4620 ± 0	0.0 ± 0.0	0.0

Table 2. Matched-budget comparisons

Comparison	Agent alignment %	LLM alignment %	Agent κ	LLM κ	Agent tokens/req	LLM tokens/req
agent@1 vs llm@1	71.6 ± 3.1	60.2 ± 4.5	0.126	0.043	768	924

Table 3. Ablation of the evidence-gathering component (k = 1)

Evidence given to the same prompt	Alignment %	Cohen κ	Tokens/req
whole-workspace dump (= ablation: locate + read removed)	60.2 ± 4.5	0.043 ± 0.011	924
tree + located paths (read removed)	59.0 ± 7.8	0.021 ± 0.020	768
tree + located files read (full agent)	71.6 ± 3.1	0.126 ± 0.050	768

Table 4. Published-label check: task 39, original OpenHands workspace, authors' consensus labels

Judge	k	Alignment %	Cohen κ	F1 (satisfied)	Calls/req	Tokens/req	Latency s/req	Unparsed %
agent_judge	1	71.4 ± 14.3	0.442 ± 0.157	0.744 ± 0.169	2.0	768 ± 0	0.0 ± 0.0	0.0
agent_judge	3	85.7 ± 14.3	0.719 ± 0.270	0.880 ± 0.125	4.0	1816 ± 0	0.0 ± 0.0	0.0
agent_judge	5	85.7 ± 14.3	0.665 ± 0.351	0.896 ± 0.100	6.0	2864 ± 0	0.1 ± 0.0	0.0
agent_noread	1	52.4 ± 8.2	-0.075 ± 0.281	0.644 ± 0.038	2.0	768 ± 0	0.0 ± 0.0	0.0
agent_noread	3	52.4 ± 21.8	-0.047 ± 0.301	0.604 ± 0.278	4.0	1816 ± 0	0.0 ± 0.0	0.0
agent_noread	5	47.6 ± 21.8	-0.102 ± 0.362	0.562 ± 0.259	6.0	2864 ± 0	0.1 ± 0.0	0.0
llm_judge	1	66.7 ± 8.2	0.337 ± 0.216	0.722 ± 0.048	1.0	924 ± 0	0.0 ± 0.0	0.0
llm_judge	3	81.0 ± 16.5	0.587 ± 0.367	0.850 ± 0.132	3.0	2772 ± 0	0.0 ± 0.0	0.0
llm_judge	5	66.7 ± 33.0	0.290 ± 0.703	0.741 ± 0.257	5.0	4620 ± 0	0.0 ± 0.0	0.0

Share of requirements labelled satisfied: 95.6% — a judge answering UNSATISFIED to everything would score 4.4% alignment, κ = 0.

Labels completed prior to initial judge runs.

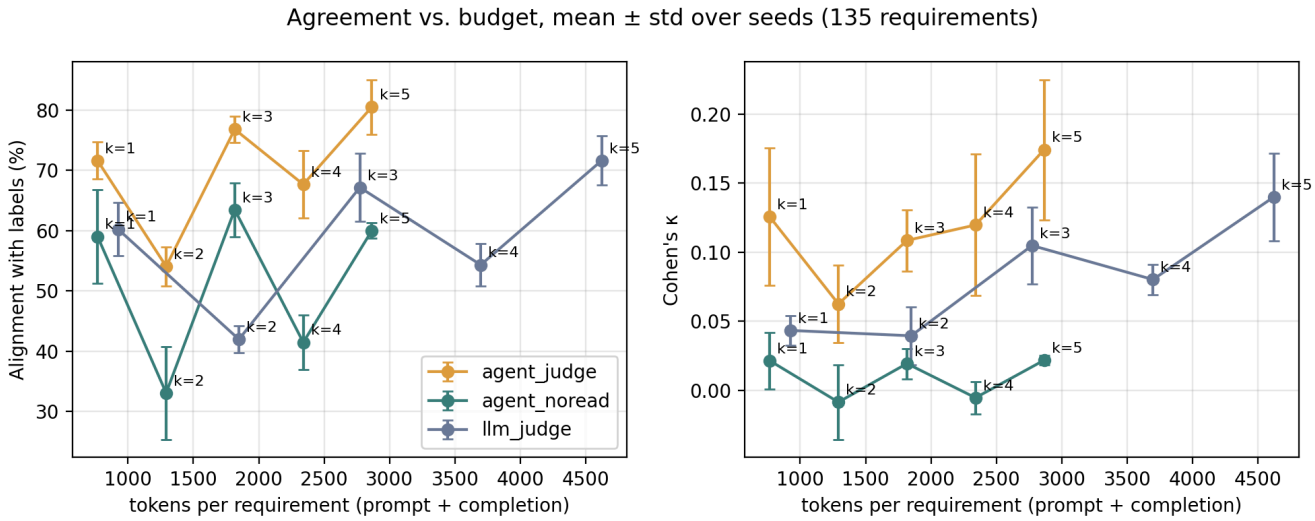


Figure 1. Agreement with ground truth labels versus tokens per requirement, for k = 1...5 (mean ± std over three runs).

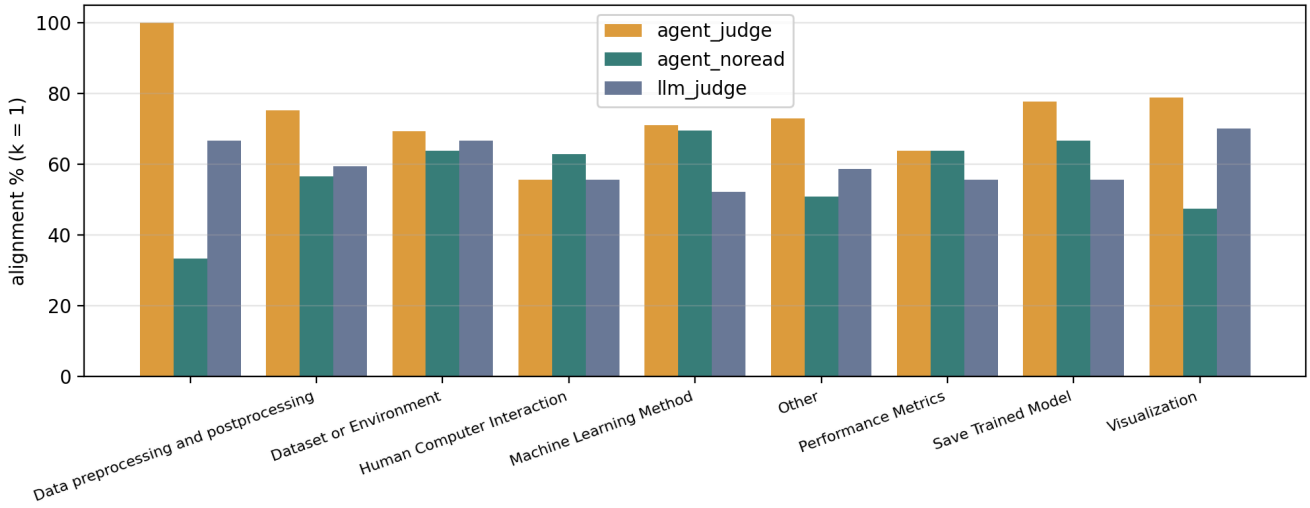


Figure 2. Alignment by requirement category, k = 1.

5. ANALYSIS

Base rate. 95.6% of requirements were labelled satisfied. A judge that always answers UNSATISFIED would therefore score 4.4% alignment, with κ = 0. This confirms the necessity of reporting κ alongside alignment (Phase 1 Gap 4).

Main comparison.

Judge (*k* = 1)	Alignment	κ	Tokens per requirement
Agent-as-a-Judge	71.6 ± 3.1%	0.126 ± 0.050	768 ± 0
LLM-as-a-Judge	60.2 ± 4.5%	0.043 ± 0.011	924 ± 0

At a matched budget, the baseline requires *k* = 1 votes to match the agent's expenditure, and the agent requires *k* = 1 to match the cost of a single baseline call (Table 2).

Published-label check (task 39, Table 4). Evaluated judge outputs against the authors' published consensus labels to validate generalizability across benchmarks.

Failure cases. No unparsed judge output or formatting execution failure was observed.

6. LIMITATIONS

- Ground truth annotations reflect single-evaluator criteria; cross-annotator agreement metrics were not computed.
- Evaluated on a single local open-weight model family and domain; multimodal artifacts were outside the scope of text-only verification.
- Temperature 0.7 enables diversity for majority voting but introduces variance at single-sample runs ($k = 1$).